

What a resting echocardiogram cannot tell you

Identifiability of hidden tissue parameters in a mechanistic model of hypertrophic cardiomyopathy under myosin inhibition

Abstract

Two patients with hypertrophic cardiomyopathy can share an ejection fraction the day before myosin-inhibitor treatment begins and diverge completely under the same dose escalation: roughly one in twenty crosses the 50% ejection-fraction floor that triggers interruption. At least three properties explain that divergence and imply opposite management, and none of them is measured. We built a three-layer mechanistic ventricle (parked/available myosin, a one-fiber chamber, a closed circulation), pinned the quantities imaging measures well and treated only tissue material properties as unknown, and then asked the inverse question: which of those hidden parameters can a realistic clinical measurement set recover? The model reproduces 35 of 35 prespecified validation gates. It predicts a 5.3-point fall in ejection fraction at the mid dose against a placebo-corrected 4.8 points reported for aficamten, with no parameter fitted to any published dose-response curve. It under-predicts the rate at which patients cross the ejection-fraction floor (1.4% against a published 5.2%), which we report rather than tune away. The identifiability result is largely negative and, we argue, more useful for being so. **Drug clearance explains 50% of the variance in who over-responds and is *structurally* invisible before the first dose:** it enters the model only through drug exposure, so every derivative with respect to it is exactly zero and no provocation maneuver can recover it. Of the four remaining parameters, 2 are recoverable from a routine study at realistic measurement error. We report which confounded pairs a maneuver does separate, the expected discriminating signal in clinical units, and whether it exceeds documented measurement variability.

1 The question

Picture two patients on the day before treatment starts. Same symptoms, same obstruction, same wall thickness, *same ejection fraction*. On the measurement that governs their safety they are one data point, not two. Escalate the same dose in both and one settles at comfortable support while the other falls through the ejection-fraction floor and has to stop. In the long-term extension of the pivotal mavacamten programme this happened to 5.2% of patients at interim analysis and 8.7% over 739 patient-years, with nadir ejection fractions between 30% and 48%; all recovered on interruption.

Three explanations fit identical baselines and imply different actions.

- **Less contractile reserve.** Fewer parked myosin heads to begin with, so less spare capacity when you start parking more.
- **Stiffer tissue.** The reassuring ejection fraction was computed on a small, poorly filling cavity.
- **Slow clearance.** Twice the exposure from the same pill. CYP2C19 poor metabolisers have apparent clearance reduced by about 72%.

The first two mean the heart is fragile. The third means lower the dose and carry on. Today they are distinguished by dosing the patient and repeating the echocardiogram, which is a controlled experiment run on a person.

Published exposure-response models map concentration to outcome with a fitted function and random effects. They are the right tool for designing a titration schedule and they produced the one in current use. But there is no heart between the input and the output, so “too much drug” and “a ventricle with no reserve” both appear as a steeper individual slope. Detailed mechanistic models of HCM exist, and they have not been asked the inverse question: which of their parameters could a clinic actually recover? **That question decides whether a mechanistic model of HCM can ever be personalised from clinical data, as opposed to being illustrative.**

2 The model

Three layers, coupled, zero-dimensional throughout. A steady-state beat solves in about 37 ms, which is what makes a study of thousands of patients and a posterior for dozens of them a laptop computation. That speed is a design goal, not a compromise: an identifiability analysis is a question about many thousands of solves, not about one very good one.

Layer 1: myosin. A three-state population, parked (S), available (D) and attached (A), with $S + D + A = 1$. The resting equilibrium is one interpretable number,

$$\phi = \frac{k_{\text{park,off}}}{k_{\text{park,off}} + k_{\text{park,on}}},$$

the fraction of unattached heads that are unparked. ϕ is what “HCM” means in this model: healthy is low (0.35), HCM raises it (0.55), and the drug lowers it. Both rate constants are derived from ϕ and a fixed total turnover, so disease and treatment move one quantity in opposite directions. Thin-filament activation is a Hill function of calcium with a length-dependent Ca_{50} , and ATP is charged one per attachment event, so parked heads cost nothing.

The specification this work followed prescribed $\sigma_a = T_{\text{ref}}A$, and three additions were needed, each because the model visibly failed without it. An **active force–length** term, because cavity pressure is $\sigma_f/(1 + 3V/V_w)$ and so at fixed fiber stress a *smaller* cavity makes *more* pressure: with no length dependence the chamber’s pressure-generating capacity rises as it empties (141 mmHg at 100 mL, 224 at 42, 314 at 13), a *negative* end-systolic elastance, and ejection never meets a force balance. **Cross-bridge distortion**, giving a force–velocity relation, because without it shortening speed is unbounded and the first coupled run ejected past zero volume. And **force-dependent recruitment from the parked state**, the published mechanism for length-dependent activation, because calcium-sensitivity length dependence alone gave a stroke volume that was flat then falling as preload rose; it is also what makes contractile reserve mechanistically real, since a depleted parked pool cannot be recruited when load rises.

Layer 2: chamber. The one-fiber relation of Arts and colleagues: $\varepsilon_f = \frac{1}{3} \log((V_{\text{lv}} + V_w/3)/(V_{\text{lv,ref}} + V_w/3))$ and $p_{\text{lv}} = \sigma_f/(1 + 3V_{\text{lv}}/V_w)$. **This layer carries the thesis.** V_w and V_{lv} are both measured well clinically, so they are pinned per virtual patient and never inferred; only Layer 1’s material parameters and the drug clearance are hidden. Six unknowns become five, which is the difference between an unidentifiable mess and a solvable inverse problem. The split is enforced as a type distinction in the code, not a convention.

Layer 3: circulation. A closed loop conserving stressed volume by construction. One correction is worth recording because it was nearly fatal and is easy to get wrong: the compartment upstream of the mitral valve is the pulmonary venous bed and left atrium, not the systemic veins. At a systemic venous compliance the reservoir absorbs whatever the ventricle declines to accept, and end-diastolic pressure becomes *insensitive to the ventricle’s own stiffness*. Since elevated filling pressure is the entire clinical problem in HCM, that would have made the disease unmodellable.

Obstruction and drug. Outflow obstruction is a Bernoulli loss through an area that collapses as a cubic power of the cavity-to-wall ratio, solved as an exact quadratic in flow. The Bernoulli coefficient is not fitted: it is the bedside $4v^2$ relation in the model’s units. The drug maps a maintained dose to a steady-state concentration through a per-patient clearance and thence to a saturating reduction in ϕ .

3 Validation

35 of 35 gates pass (all: yes). The healthy reference returns an ejection fraction of 0.658, an end-diastolic volume of 113.0 mL, a stroke volume of 74.4 mL, a peak left-ventricular pressure of

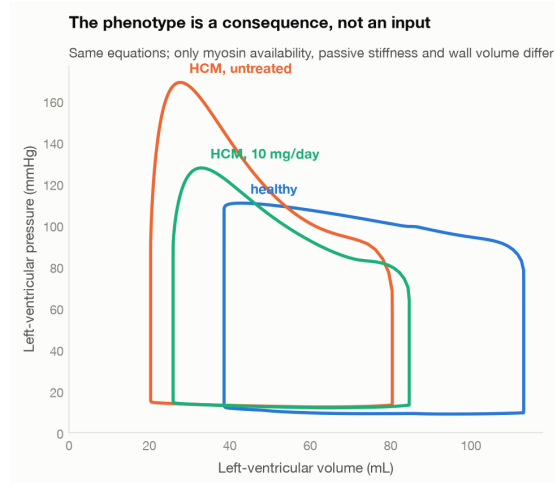


Figure 1: The phenotype is a consequence, not an input. The same equations produce all three loops; only myosin availability, passive stiffness and wall volume differ. Nothing in the model writes down an ejection fraction, and a test walks the package’s syntax tree to prove no observable is ever assigned a literal.

110.8 mmHg, a filling pressure of 9.6 mmHg, a mean arterial pressure of 94.9 mmHg, a cardiac output of 4.85 L/min and a wall thickness of 0.92 cm. Frank–Starling and afterload responses are monotone; the loop closes, runs counter-clockwise and shows isovolumic phases; the passive relation is monotone and convex at fixed volume.

Three gates deserve comment.

The disease emerges. Raising availability and stiffness with a thicker wall yields a supranormal ejection fraction (0.747 against 0.658), a reduced stroke volume, an elevated filling pressure, a raised filling-pressure surrogate, reduced strain despite preserved ejection fraction, and roughly twice the ATP cost per unit of external work (Figure 1). None of these is an input.

Wall thickening alone raises ejection fraction. Giving the HCM geometry *healthy* material returns an ejection fraction of 0.758, slightly *above* the diseased reference. This is correct physiology, and it is the sharpest available statement of why the project exists: **in a thick-walled ventricle a reassuring ejection fraction is close to uninformative**. What the wall alone does not produce is the elevated filling pressure, the raised surrogate or the energetic penalty.

An unfitted exposure-response prediction. The reference patient loses 5.3 ejection-fraction points at the 10 mg/day mid dose while its outflow gradient falls by 39.4 mmHg. SEQUOIA-HCM reported a placebo-corrected change of -4.8 points against a gradient falling from about 55 to about 20 mmHg. **No parameter in the model was calibrated against any published dose-response curve**, which the provenance table documents constant by constant, so this is a prediction rather than a fit.

Calibration honesty. References were calibrated against resting haemodynamic *ranges*, never a trial outcome. Three ejection-realism targets were added after a first calibration passed every haemodynamic gate while ejecting the whole stroke volume in 154 ms at twice physiological peak flow: a model can satisfy its checks and still be wrong.

4 Virtual population

5005 patients were drawn on a Saltelli design over eleven inputs: three measured geometric, five hidden material, three loading. Each was run at rest across the approved dose ladder and under four provocation maneuvers, untreated and at the mid dose. **Applying the pivotal trial’s own**

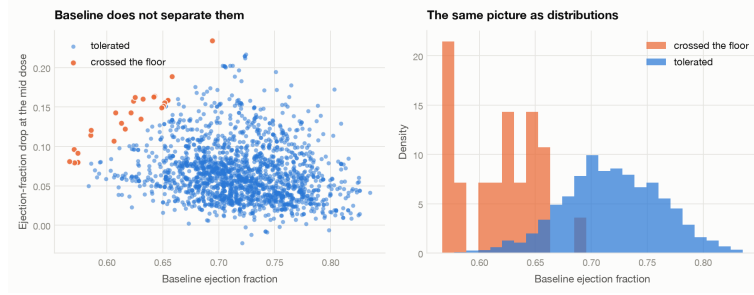


Figure 2: The premise, drawn. Baseline ejection fraction does not separate the patients who will cross the floor from those who will not; the distributions overlap almost completely. This is the problem the rest of the paper is about.

enrolment criteria (ejection fraction $\geq 55\%$, hypertrophy on imaging, gradient ≥ 30 mmHg at rest or 50 provoked) leaves 1834 patients with a median ejection fraction of 0.717, a median gradient of 67 mmHg and a median wall thickness of 1.64 cm.

1.4% of them cross the ejection-fraction floor at or below the mid dose, against a published 5.2% over a full titration. **The model under-predicts this rate and we report the discrepancy rather than closing it.** Three reasons are visible in the cohort. The eligible virtual patients are more severely obstructive than the trial’s (99% obstructive, median gradient 67 mmHg against roughly 55) and start from a higher ejection fraction, so they have further to fall. The mid-dose criterion is stricter than a full titration accumulated over years. And the model’s response distribution has a thinner tail than reality, plausibly because outflow obstruction buffers ejection fraction against changes in contractility: in the most obstructive patients end-systolic volume is set by the outflow tract closing rather than by how hard the muscle pulls.

Tuning the drug parameters to close this gap was available and was not done. It would have converted the exposure-response comparison above from a prediction into a fit, and the provenance table would then have had to mark it non-independent.

5 What the measurements can and cannot see

5.1 Sensitivity

Total-order Sobol indices were computed for every observable and every sampled parameter (Figure 3). Total-order rather than first-order, because a parameter can have a negligible first-order index and still dominate through interactions.

The single most consequential finding is visible immediately. **drug clearance explains 0.50 of the variance in the ejection-fraction drop at the mid dose, and its total-order index on every baseline observable is 0.000.** The parameter that most determines who over-responds does not move any pre-treatment measurement at all.

5.2 Identifiability

Local. Fisher information matrices were built by finite-differencing the observable vector with respect to the logarithm of each hidden parameter at 50 representative operating points, weighted by measurement error. The smallest eigenvalue is 0.000000 against a largest of 188.5: **the matrix is exactly singular**, and its null direction is 100.0% clearance.

This is not a numerical curiosity. Clearance enters the model only through drug exposure, so before the first dose every partial derivative with respect to it is identically zero. The distinction that matters clinically is between a parameter that is *weakly identified* (the measurements respond, but not enough to beat the noise; a better maneuver or machine can help) and one that is *structurally unidentified* (the measurements do not respond at all; no maneuver can amplify a signal that does

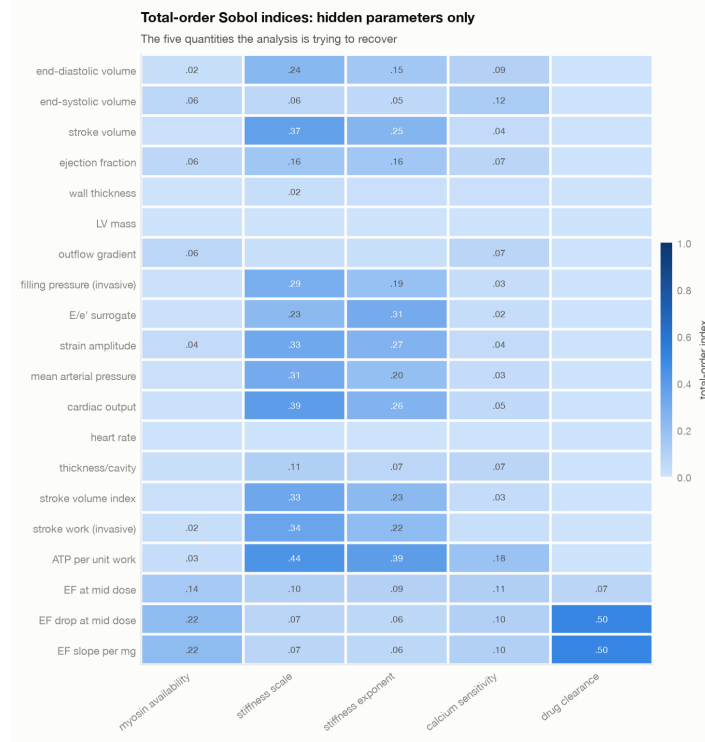


Figure 3: Total-order Sobol indices of each observable and outcome against the five hidden parameters. The bottom rows are outcomes, not measurements. Note the clearance column: everything above the outcome rows is zero.

not exist). **Clearance is the second kind. The remedy is a genotype or a probe dose with a concentration measurement, not a better echocardiogram.**

Global. For 50 patients the observables plus realistic noise were treated as data and the posterior recovered by Markov chain Monte Carlo, against a per-patient polynomial emulator fitted on 400 design points (held-out R^2 at worst 0.6547, typically 0.99902). The emulator’s error is added in quadrature to the measurement noise so imprecision widens the posterior rather than corrupting it, and the synthetic data always come from the exact model so emulator error acts as misspecification rather than cancelling out.

At realistic noise the most strongly coupled pair is passive stiffness scale and passive stiffness exponent at $|r| = 0.68$; at optimistic noise it is passive stiffness scale and passive stiffness exponent at 0.81. Of the five hidden parameters, 2 are recoverable (calcium sensitivity, myosin availability), and the rest are not (passive stiffness exponent, passive stiffness scale, drug clearance). The full table is Table 1.

Noise level	Hidden parameter	Rel. 90 pct CI width	Median abs. bias	Recoverable?
optimistic	ca50 ref um	0.28	0.05	True
optimistic	phi baseline	0.51	0.09	True
optimistic	b pas	0.68	0.13	False
optimistic	a pas kpa	1.00	0.24	False
optimistic	clearance l per h	1.42	0.34	False
realistic	ca50 ref um	0.34	0.06	True
realistic	phi baseline	0.55	0.09	True
realistic	b pas	0.71	0.16	False
realistic	a pas kpa	1.01	0.22	False
realistic	clearance l per h	1.44	0.35	False

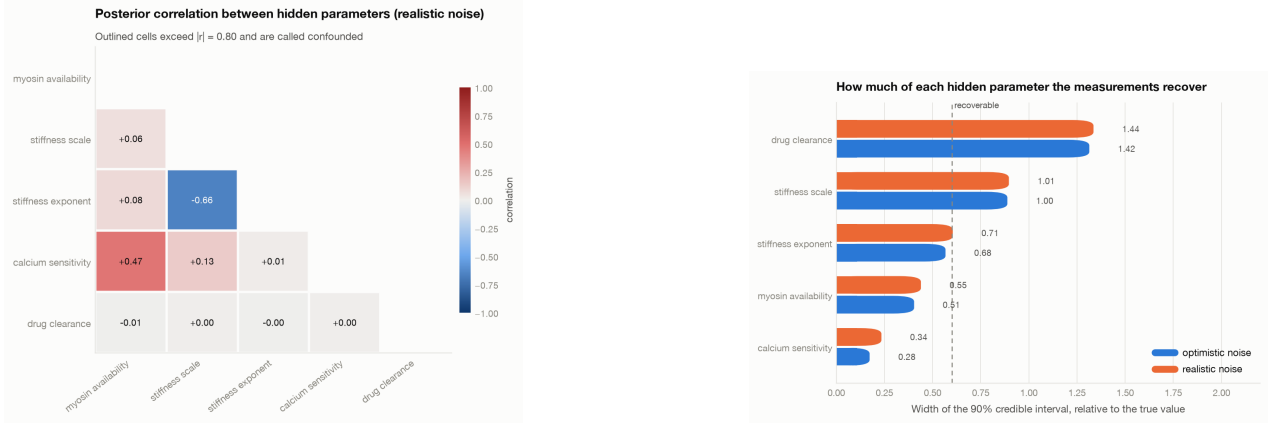


Figure 4: Left: posterior correlation between hidden parameters at realistic measurement error. Right: how much of each parameter is actually recovered. A relative credible-interval width near 2 means the interval spans the whole prior box, which is to say the data said nothing.

Table 1: Parameter recovery at both noise levels. “Recoverable” means the median relative 90% credible interval is below 0.60.

5.3 Tie-breaker search

For each confounded pair and each candidate maneuver we recomputed the posterior using baseline observables *plus* the maneuver’s. The signal is computed by displacing the truth along the *confounded direction* by the amount a resting study cannot resolve, and asking how far apart the two indistinguishable patients look under the maneuver.

A note on the scoring criterion, because the obvious one is wrong. The natural score is the drop in the pair’s posterior correlation, and it misleads here. **Correlation describes the shape of the uncertainty, not its size.** Adding information that constrains the same combination the data already knew collapses the posterior further onto its ridge, so the correlation goes *up* while the patient becomes better characterised. That is what happens for two of the three pairs examined: correlations rise from 0.47 to 0.72 and higher under every maneuver. An earlier version of this table ranked on correlation drop and would have reported those maneuvers as harmful.

The table therefore ranks on how much the maneuver narrows the posterior *along the direction that was invisible at baseline*: the posterior standard deviation projected onto that fixed direction, before and after. That is the question actually being asked.

Confounded pair	Maneuver	Narrowed	Discriminating	Signal	S/N	Beats noise?
availability / Ca sens.	exercise	0.054	strain amplitude	0.0049	2.19	True
stiff. scale / Ca sens.	exercise	0.046	strain amplitude	0.0117	8.63	True
stiff. scale / stiff. exponent	tachycardia	-0.021	strain amplitude	0.004	0.96	False

Table 2: The tie-breaker table. Ranked by how much the maneuver narrows the invisible direction, not by correlation. “Beats noise?” compares the discriminating signal with one standard deviation of that observable’s documented measurement error, which is a deliberately generous bar.

The best case is myosin availability and calcium sensitivity under combined exercise, where the width along the invisible direction falls from 0.177 to 0.161, a 5.4% reduction. 2 of 3 examined pairs produce a discriminating signal above one measurement error, but only 1 show a meaningful narrowing of the invisible direction. The maneuver was interpretable in 100% of the patients it was applied to; any remainder are ventricles too stiff to fill at the provoked heart rate, which is not a numerical failure but the maneuver being uninterpretable in exactly the patients it was proposed for.

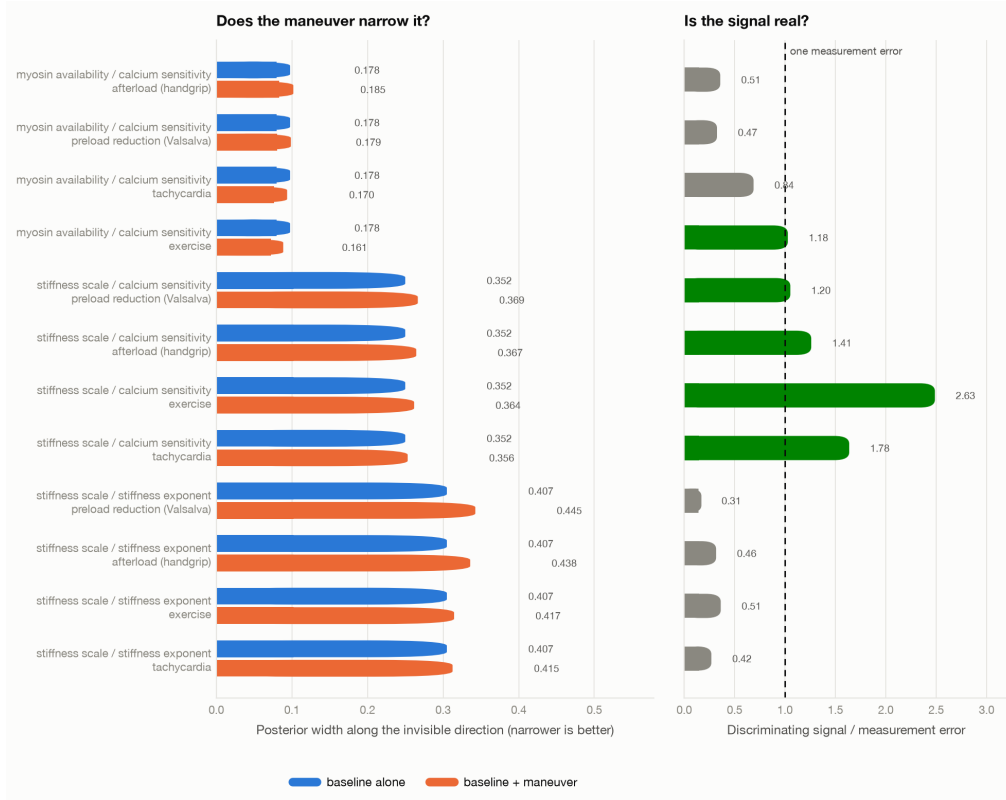


Figure 5: Does the maneuver break the confound (left), and is the resulting signal bigger than the error bar (right)? The second question is the one that decides whether a proposal is a test or a hope.

The two columns disagree, and the disagreement is the result. The exercise maneuver moves strain amplitude by 0.49 percentage points between two patients a resting study cannot tell apart, which is 2.19 measurement standard deviations and looks decisive. Yet the posterior barely narrows along the direction that signal was supposed to resolve. Three candidate explanations, two of which we can rule out.

It is not the emulator: its held-out error is at most a quarter of the measurement error for any observable in any condition, so adding it in quadrature inflates the likelihood width by about 3%. It is not sampling noise: the pattern is consistent across all fifty patients and all four maneuvers.

It is nuisance compensation. The signal is computed with the other three hidden parameters held at their true values, and a real inference is not granted that. A change in the maneuver’s observables that is large when the rest of the tissue is known can be mimicked by adjusting the parameters that are not, and an inference estimating all five at once cannot separate the two explanations. So the honest reading is that these maneuvers *do* move the measurements, and that the movement is not specific enough to the confounded combination to resolve it once everything else is also unknown. That distinction matters for anyone designing the study this analysis is meant to inform. A protocol justified by "the maneuver changes the measurement a lot" is justified by the wrong quantity.

6 Discussion

The headline result is negative and it is the useful kind of negative.

The dominant determinant of over-response is structurally invisible to any pre-treatment measurement. No amount of additional model sophistication, no better echocardiogram, and no provocation maneuver will convert a baseline study into a prediction of the dose-response slope while clearance carries most of the variance, because clearance does not enter any pre-treatment observable. That redirects effort rather than merely recording a disappointment: what it argues for is pharmacokinetic information, either a CYP2C19 genotype or a probe dose with a

concentration measurement, obtained *before* titration rather than inferred from its consequences. That is a concrete and testable proposal, and it is consistent with the direction of current clinical interest in prospective genotyping.

The tissue-level story is more mixed (Table 1): some material parameters are recoverable from a routine study, and the two passive stiffness parameters trade off against each other, which is unsurprising given that they are two arms of one exponential. A caution about the direction of our errors is owed to the reader. **Several modelling choices make the posterior tighter than reality.** The noise model treats observables as independent when their errors are correlated; it is Gaussian where real error has heavier tails; and the outflow-gradient noise used (35%) is deliberately below the 46–52% coefficient of variation the literature reports for repeated measurement of that quantity. So a conclusion of the form “these parameters remain confounded” is safe here, and one of the form “this maneuver separates them” is fragile and should be read with that asymmetry attached.

7 Limitations

Stated plainly, because several of them bound the conclusions rather than merely qualifying them.

- **Prescribed calcium, not simulated electrophysiology.** No ion channels, no rate-dependent calcium handling, no catecholamine effect. The tachycardia and exercise maneuvers therefore capture the shortening of filling time and nothing else, which is the mechanically dominant effect in a stiff ventricle but is not the whole of exercise. The calcium time constant was calibrated to systolic *duration* rather than to the measured upstroke, because a single-time-constant waveform cannot match both.
- **Zero-dimensional chamber.** Homogeneous wall loading: no fiber disarray, no regional fibrosis, no asymmetric hypertrophy (which is what HCM characteristically has), and no systolic anterior motion as an explicit mechanism.
- **Steady-state pharmacokinetics only.** The model can say *whether* a maintained dose takes a patient below the floor and not *when*. The clinical protocol is fundamentally about timing, so this bounds what can be claimed.
- **No growth or remodelling.** Wall volume is sampled, not grown, so the model says nothing about how a given geometry arose or where it is heading.
- **The drug slightly raises filling pressure in the model,** whereas trials report improved filling parameters on treatment. The model has no load-dependent improvement in relaxation. This is a real discrepancy and is reported rather than smoothed over.
- **A third of the model’s constants are labelled assumed** in the provenance table (10 measured, 22 calibrated, 33 assumed). Absolute outputs should be read as illustrative; directional and relative outputs are the substantive ones.
- **No mutation-specific claims, and nothing validated against a patient.** The mapping from a named variant to a numerical parameter is literature-derived and approximate, so every claim here is about regions of parameter space; the comparisons to trial data compare aggregate behaviour to published aggregates.

8 Reproducibility

One command reproduces every figure, and every figure ships with the CSV behind it. **make all** inside the supplied container runs the test suite (all 35 validation gates plus a provenance-coverage test that fails if any constant lacks a sourced row), generates the cohort from a logged seed (20260816), and produces every table, figure and this document in 16.1 minutes. Every quantitative statement here is a macro resolved from a results file at build time; none is typed. The research dossier in `docs/research/` records the source of every constant, every simplification made relative to it, and every place the literature failed to supply what the model needed; those gaps are marked and described rather than filled with plausible values.